

A Tiered Roadmap for Calculations in Modal Triplet Theory (MTT): Audited Closure and Strict Upgrades

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Abstract

We replace the former execution-complete roadmap with an audited hierarchy of closure standards. MTT currently closes embedded renormalized-Standard-Model equivalence at the adopted one-shared-physical-primitive/measured-profile standard: the final audit passes 12/12 obligations using selected multi-loop transport, the 27-by-27 operator architecture, charged flavor and CKM rows, Higgs/threshold support, and imported standard SM quantization. This is not a zero-knob derivation or unique observed-branch theorem. The obsolete few-TeV crossing, Tier-3 backfit, Calabi–Yau numerical lift, fitted flavor matrices, and cosmological tensor bound are retired. We list the exact surviving structural theorems, conditional relations, profile closures, and nine strict upgrades, of which U2 and U4 are closed. The roadmap now measures progress by named source objects and reproducible acceptance tests rather than repeated status declarations.

Revision note for this edition

Supersedes. *A Tiered Roadmap for Calculations in Modal Triplet Theory*, version 2.

Reason. The former roadmap mixed obsolete benchmark calculations with open source theorems and repeatedly reopened already certified matrix, Yukawa, electroweak, and threshold work.

Resolution. Version 3 locks the 12/12 embedded-renormalized-SM baseline, retires the few-TeV/CY backfit chain, and replaces repeated status targets by nine named strict-upgrade objects and acceptance tests.

Retained result. The current one-shared-physical-primitive/profile closure and its reproducible packets remain the operational baseline.

Remaining boundary. Zero-knob and unique-branch selection are stronger upgrades and must not be confused with reopening the closed baseline.

1 Closure vocabulary

Every result must carry one of the following labels:

Structural theorem. Derived from selected mathematical source data.

Conditional theorem. Derived only after all displayed assumptions are imposed.

Calibration. Parameters are chosen using the compared data.

Profile/replay. Measured coordinates are transported or embedded in a declared scheme.

Diagnostic. A consistency check whose target entered selection or fit.

Held-out prediction. The target is absent from source selection, calibration, branch choice, and uncertainty construction.

No result may move between these classes merely because its numerical residual is small.

2 Current headline theorem

The permitted headline is:

MTT closes embedded renormalized-Standard-Model equivalence at the adopted one-shared-physical-primitive/profile standard on the selected branch.

It must be accompanied by the limitation:

This is an embedding/parity result with measured profile inputs and imported standard SM quantization. It is not zero-knob derivation, unique observed-branch selection, or a derivation of perturbative quantization from MTT.

The numerical authority is the SMDR v1.3 common-scheme transport at $Q = M_t = 172.5590883453979$ GeV. Fifteen source coordinates map to eight full-SM $\overline{\text{MS}}$ rows with a positive-definite 8×8 covariance, all 36 symmetric entries, and all 15 BCT–WZH cross entries.

3 Revised tier architecture

3.1 Tier 0: conventions and provenance

Tier 0 fixes units, hypercharge normalization, renormalization scheme, scale, branch identifiers, source ownership, covariance, and code versions. It produces no physical prediction. This tier is mandatory for every numerical claim.

3.2 Tier 1: exact finite and topological structure

Tier 1 contains exact results only with their branch hypotheses. Current examples include the selected $q = 79$ modulo-448 branch, finite-gerbe/charge data, the 27-by-27 qutrit–Weyl architecture, finite Cech tables, cocycle identities, and selected retarded orientation representative. These do not prove uniqueness over all MTT branches or automatically determine measured scalar magnitudes.

3.3 Tier 2: geometry-light conditional relations

Tier 2 now separates exact algebra from assumptions. Modal democracy implies $\sin^2 \theta_W = 3/8$ but is not selected by the current profile. Holonomy phase sums require a connection-preserving trivialization. Equal wave speeds require a common principal symbol. Curvature–mass and PPN statements require explicit mass ansatz, positivity, response norms, and physical normalization. Internal spectral gaps are not external cutoffs.

3.4 Tier 3: profile-standard superset execution

The legacy crossing-scale backfit is retired. Current Tier 3 is the audited superset embedding:

- final profile-standard audit: 12/12;
- one shared physical primitive P_{EW} and zero Higgs-specific parameters at the embedding layer;

- selected multi-loop transport and covariance;
- ten magnitude labels and physical threshold/mass coordinates retained transparently as profile data; and
- standard SM perturbative quantization imported rather than derived.

3.5 Tier 4: selected geometry and operator execution

The old fitted Calabi–Yau corner is not validated and is retired as numerical authority. The selected rank-two literal Cech–HYM execution is closed 2/2, including finite patching and the continuum tail/Wiener certificate. The internal $K_{\text{threshold}}$ response closes 10/10. Charged flavor and Higgs close at the profile standard; CKM also closes at the selected prediction-with-uncertainty criterion. Rank-three transfer, strict scalar value emission, absolute neutrino completion, and strong CP remain stronger targets.

4 Current result matrix

Area	Current status	Scope guard
Embedded renormalized SM	closed 12/12	measured profile and imported SM quantization
27-by-27/charged operator	closed structurally	magnitudes remain profile rows at strict tier
Literal rank-two Cech–HYM	closed 2/2	no all-branch uniqueness or rank-three transfer
Internal K threshold	closed 10/10	physical threshold values remain profile rows
CKM	U4 closed	uncertainty-profile prediction, not moving-central-value identity
Neutral response	internal dimension-less 9/9	physical unit and Dirac/Majorana completion open
Strong CP	partial	selected nonzero anomaly-matching/relaxation map open
Gravity TT carrier	scoped support closed	Newton normalization and full stress response open
Quantization/QFT	conditional/partial	MTT Born/BRST and constructive 4D limits open

5 Strict-upgrade program

The baseline is not reopened by stronger goals. The nine upgrades are:

- U1. zero-primitive empirical source: open;
- U2. literal global Cech–HYM/QaSU3: closed 2/2;
- U3. official joint input likelihood: partial;
- U4. CKM prediction beyond source rows: closed at profile criterion;
- U5. absolute neutrino mass and ontology: partial;
- U6. strong-CP selection: partial;
- U7. MTT-derived quantization: partial;
- U8. constructive nonperturbative four-dimensional QFT: partial; and

U9. unique observed-branch selection: partial at orientation level.

6 Next computational milestones

1. **Neutral physical completion.** Select either a non-affine spectral-action slope plus one universal scale, or dimensionful Dirac–Majorana/seesaw blocks. Common scaling alone is proved impossible.
2. **Strong CP.** Construct the selected flux/threshold axion-current map that changes the cancelling complete-27 anomaly into a justified low- energy effective anomaly, then test quality and EDM bounds.
3. **Strict scalar source.** Emit the nine charged magnitudes, λ_H , and physical threshold/mass rows from selected source data, or retain them permanently as declared profile coordinates.
4. **Branch uniqueness.** Upgrade the selected retarded orientation representative to a global action/measure and carrier uniqueness theorem.
5. **Foundational QFT.** Derive Born/record and gauge-orbit rules and construct the required nonperturbative infinite-volume four-dimensional limit.
6. **Zero-primitive synthesis.** Attempt U1 only after its dependency rows close; do not replace missing source values with another calibration.

7 Reproducibility contract

Every milestone must ship raw inputs, immutable source identifiers, units, scheme/scale/loop order, branch choices, parameter ledger, covariance, executable code, deterministic outputs, unit tests, and an acceptance certificate. Held-out claims additionally require a data-separation certificate. Historical packets remain provenance and cannot override their audited successors.

8 Conclusion

The program has advanced substantially beyond the original roadmap, but not in the manner that document claimed. The strongest current achievement is a reproducible, shared-operator embedding of the renormalized Standard Model at a clearly declared profile standard, together with exact finite/HYM and CKM strict upgrades. The old Tier-3 and Calabi–Yau numerical chain is retired.

Progress toward a deeper theory is now sharply measurable: close U5, U6, U7–U9 and finally U1 with source theorems rather than fitted replacements. This roadmap is intentionally non-looping: a row changes status only when its named missing object passes its stated acceptance test.

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